

Polytopes

A **polytope** is a bounded polyhedron.

- What do the face posets of polytopes look like?
- What do the face numbers of polytopes look like?

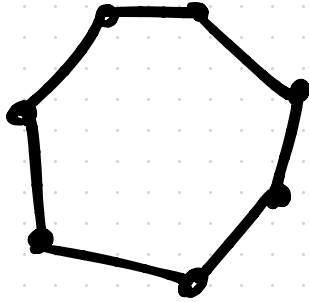
The boundary of a d -dim polytope (d -polytope)
is a $(d-1)$ -sphere.

A **polytopal complex** is a topological space formed by gluing together polytopes (**cells**) face-to-face.

We will assume pairs of polytopes are glued along at most one face, and polytopes are not glued to themselves.

A simplicial complex is a polytopal complex whose cells are simplices.

2-polytopes (1-spheres):



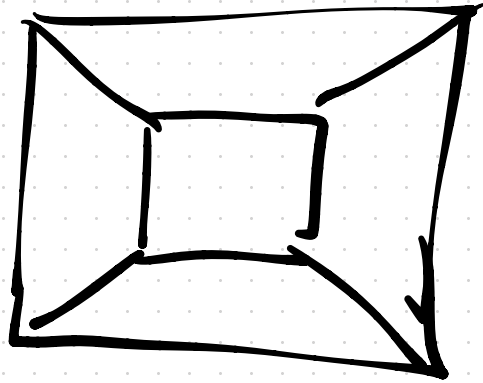
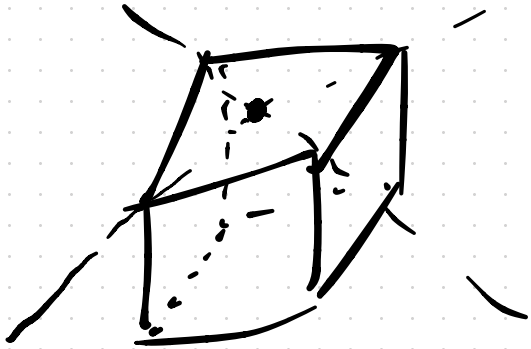
n vertices, n edges,
connected in a cycle.

Polytopes and polyhedral complexes
are the same.

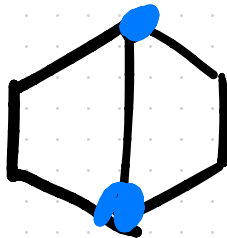
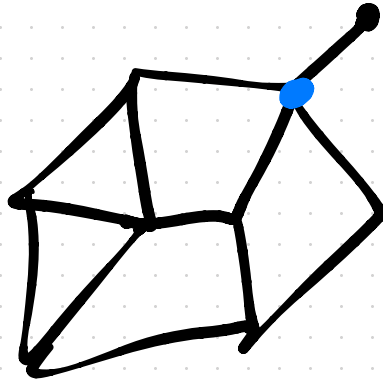
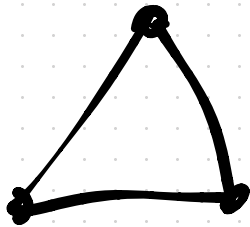
3-polytopes

(0 and 1-dim faces)

The 1-skeleton of a 3-polytope is a planar graph.



Not every plane graph is the 1-skeleton
of a 3-polytope.



A graph is k -connected if it has at least $k+1$ vertices, and deleting any $k-1$ vertices results in a connected graph.

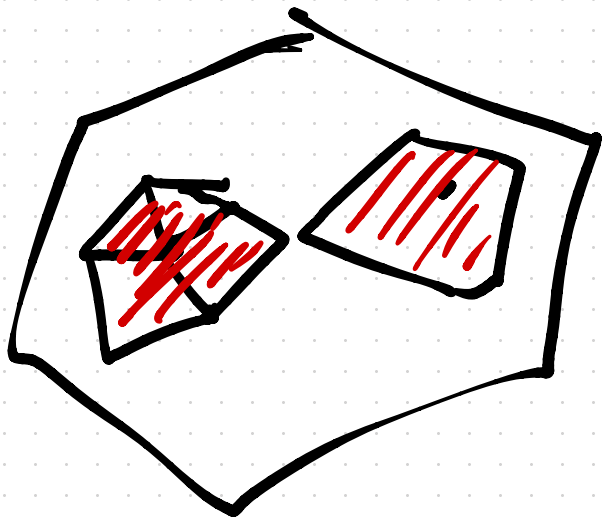
Thm (Steinitz): A graph is the 1-skeleton of a 3-polytope iff it is planar and 3-connected.

Thm (Whitney): Given a 3-connected planar graph G , every planar embedding of G has the same face structure.

Prop: The 1-skeleton of a 3-polytope is planar and 3-connected.

Thm (Balinski): The 1-skeleton of a d -polytope is d -connected.

→ Proof: Assume there are two vertices whose deletion disconnects the 1-skeleton.



This means that deleting the open stars of these vertices in the boundary of the polytope connects the boundary.

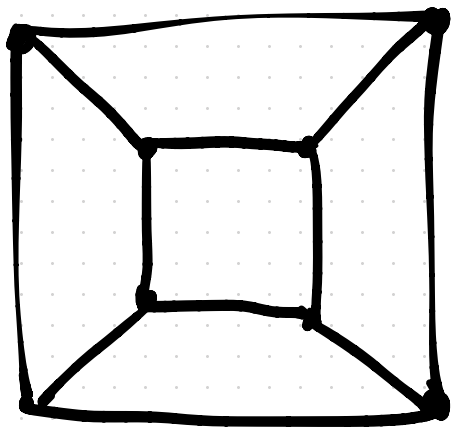
The open stars of vertices in a polytope "act like convex sets", in that the intersection of any of them is contractible. So the union of these open stars is either one or two open balls,

which cannot disconnect a sphere.

Thm: If a graph is planar and 3-connected, then it is the 1-skeleton of some 3-polytope.

Thm (Tutte): If a graph is " " , and F is one of its faces, then it has a planar embedding s.t.

- F is the outer face, and its boundary is a convex polygon.
- Every edge is a line segment
- Every interior vertex is the average of its neighbors.
- Every interior face is a convex polygon.



To find this embedding.

- Embed F as any convex polygon.
- For each internal vertex v , we create two variable x_v, y_v

- For each v , we need

$$\sum_{u \in S(v)} (x_u - x_v) = 0$$

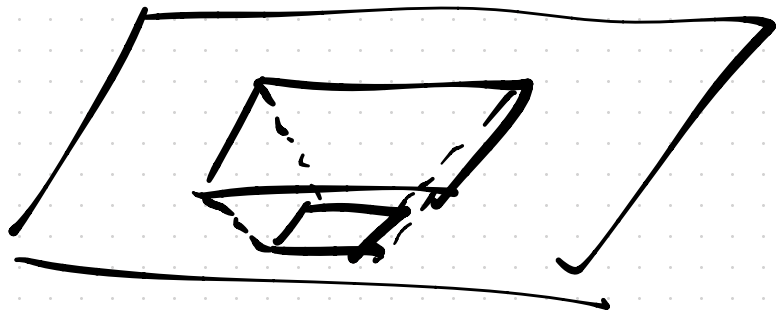
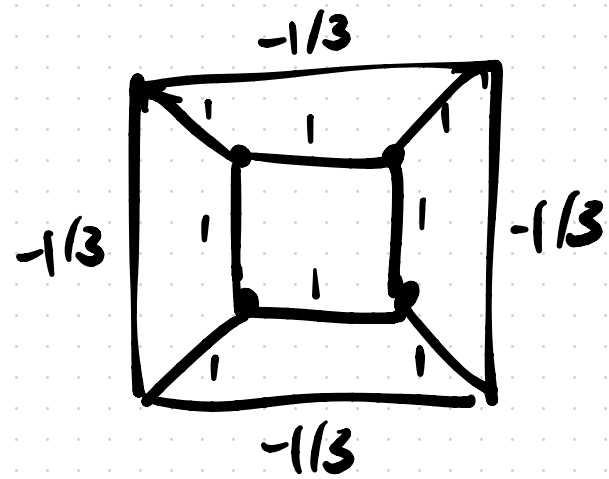
$$\sum_{u \in S(v)} (y_u - y_v) = 0$$

↑
neighborhood of v

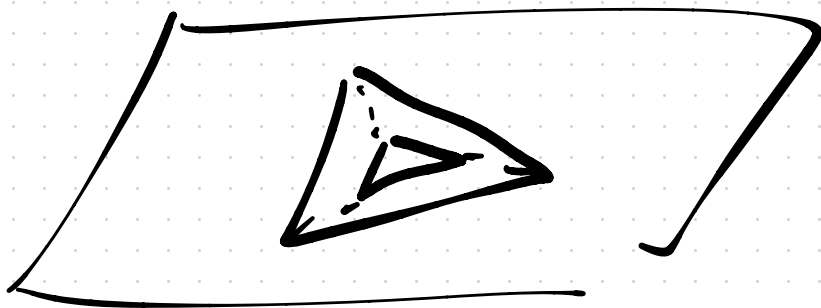
Then (Maxwell-Cremona): Let G be a straight-line embedding of a graph in the plane. Suppose we assign a real number $w(e)$ to each edge e so that for every vertex v ,

$$\sum_{u \in S(v)} w(uv)(u-v) = 0.$$

Then there is a piecewise linear function which is linear on each face of G , and which is convex (concave, linear) at each edge with positive weight (negative, 0).

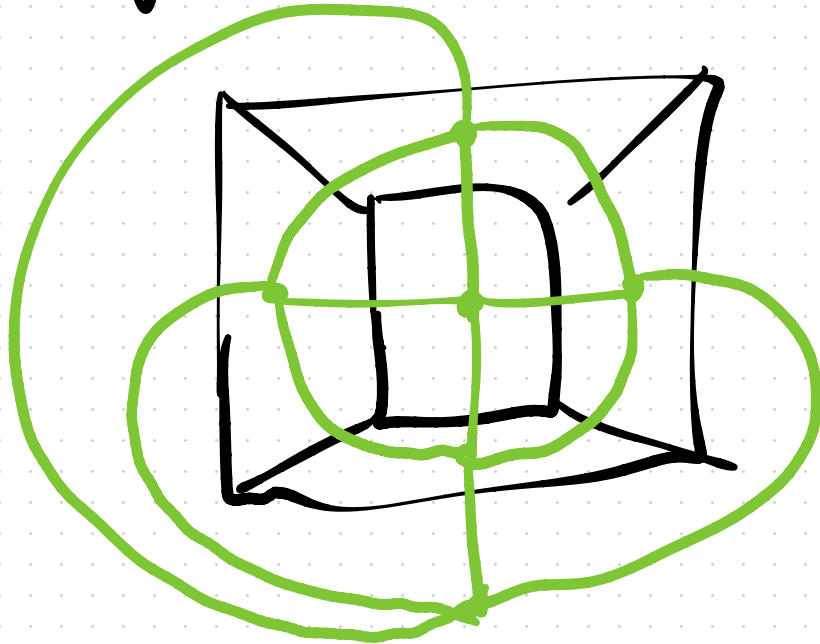


Prop: If G is a Tutte embedding and the outer face is a triangle, then it has a Maxwell-Cremona lifting where every internal edge has positive weight.



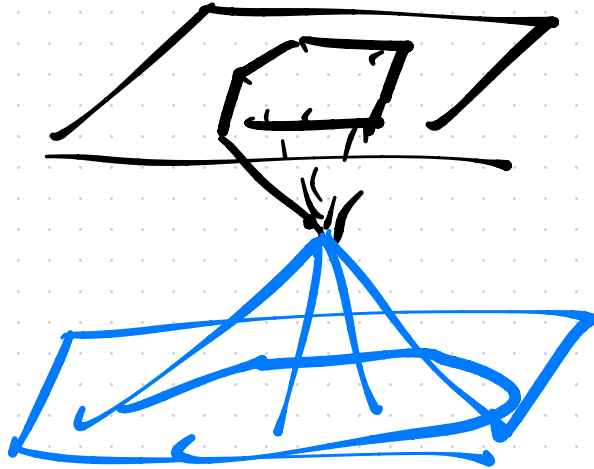
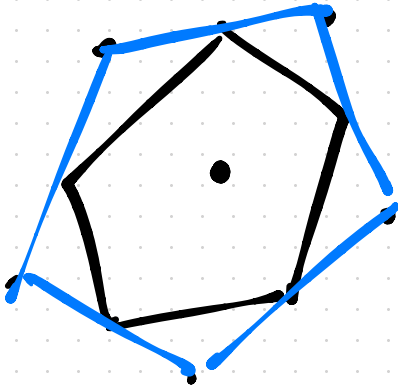
So G can be realized as the skeleton of a poly, if it has a triangular face.

Given a planar embedding G , the **dual embedding** is an embedding of a graph G^* whose vertices are the faces of G , and edges connect faces of G which share an edge.

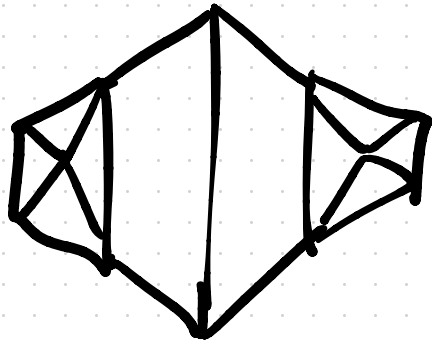


Prop: If G is 3-connected and planar, then either G or G^* has a triangular face.

Prop: If G is 3-connected and planar, then G is the 1-skeleton of a 3-polytope iff G^* is.



What about polytopal complexes which are 2-spheres?



Polytopal complex, 2-sphere

Not boundary of a 3-polytope
since it's not 3-connected

Prop: Every simplicial 2-sphere is the boundary of a 3-polytope.